

B30801 Managerial Accounting Autumn 2025

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**Session #2: Cost Behaviors;
Manufacturing Costs/Absorption Costing;
Knowledge in Organizations**

CHICAGO BOOTH



The University of Chicago Booth School of Business

RegionFly: Supply-driven Death Spiral.

- Profitability *perceived* as too low (falling margins).
- Drop products based on % profit margins calculated after allocating indirect/overhead costs.
- Volumes (revenues) decrease as we drop products.
- Overhead burden rate increases (why? overhead costs decline more slowly than VDCs used to allocate).
- % profit margins decline further, causing more products to be dropped—revenues decline more than costs.
- Cycle continues until business can no longer cover fixed costs (and so “death spiral”).

Contribution Margin Approach													
			2015		2016		2017		2018				
Revenue			6,002,529		5,713,233		4,416,922		4,164,340				
Variable Costs:													
	Variable Direct Costs		1,013,732	0.169	1,053,342	0.184	771,782	0.175	722,206	0.173			
	Flight Support (1000)		755,242	0.126	787,084	0.138	576,695	0.131	538,631	0.129			
	Maintenance (3000)		485,512	0.081	510,102	0.089	373,751	0.085	349,082	0.084			
	Airport expenses (7000)		242,756	0.040	252,165	0.044	184,761	0.042	172,566	0.041			
			2,497,242		2,602,693		1,906,989		1,782,485				
Contribution Margin			3,505,287	0.584	3,110,540	0.544	2,509,933	0.568	2,381,855	0.572			
Non-variable Costs:													
	Sales expenses (2000)		134,865	0.022	138,900	0.024	117,983	0.027	111,485	0.027			
	Financing, fees, taxes (4000)		755,242	0.126	783,004	0.137	761,080	0.172	717,969	0.172			
	Depreciation (5000)		269,729	0.045	281,473	0.049	264,022	0.060	242,362	0.058			
	Admin costs (6000)		53,946	0.009	56,287	0.010	53,585	0.012	47,592	0.011			
			1,213,782		1,259,664		1,196,670		1,119,408				
Pre-tax profit			2,291,505	0.382	1,850,876	0.324	1,313,263	0.297	1,262,447	0.303			

Is contribution margin (in \$) large enough to cover fixed costs?
Do products (routes) provide incremental contribution margin?

RegionFly: Key takeaways

- Conclusion may seem obvious here, but harder to detect in real-world situations with many products, services, and product lines.
- What metric do we ultimately care about?
 - % operating margins or aggregate \$ of contribution to bottom-line? (free cash flow or ROIC).
 - For any entity to survive in the long-run it needs to generate free cash flow.
 - *CFOs focus on EBITDA \$ and cash flow conversion.*

MA Toolbox/key ideas

- Rules of the Game: Overarching framework.
- Fixed costs, variable costs (and importance of the “relevant range”)
- Contribution margin approach.
- Opportunity costs, sunk costs.
- Cost allocation – how and why. Manufacturing costs and absorption costing.

Overarching focus: provide information useful for management decisions and control.

Why Costs and Cost Behavior Matters:

- Control – we need to understand and manage costs, esp. as business scales up (or down, as in COVID) and now in an inflationary environment including effects of tariffs.
- Planning and Solvency – we need to understand cash flows and financing – can we pay our bills?
- Deals and Valuation – we need to forecast future margins, profitability, and free cash flow, both internally and externally (for investors, analysts, lenders, etc.). Key element of DCF/NPV.

Companies develop long-term plans and “aspirational” goals under which bottom-line EPS is expected to grow faster than the top-line.

Companies use a variety of levers to achieve this (e.g., financial engineering via capital and tax structuring) but **cost management** to drive **operating leverage** (COGS and SG&A % decline as the business scales) is key.

How profitability changes as business scales up or down

Starbucks' Investor Day, Sept. 13, 2022:
Non-GAAP EPS growth for FY23 through FY25.

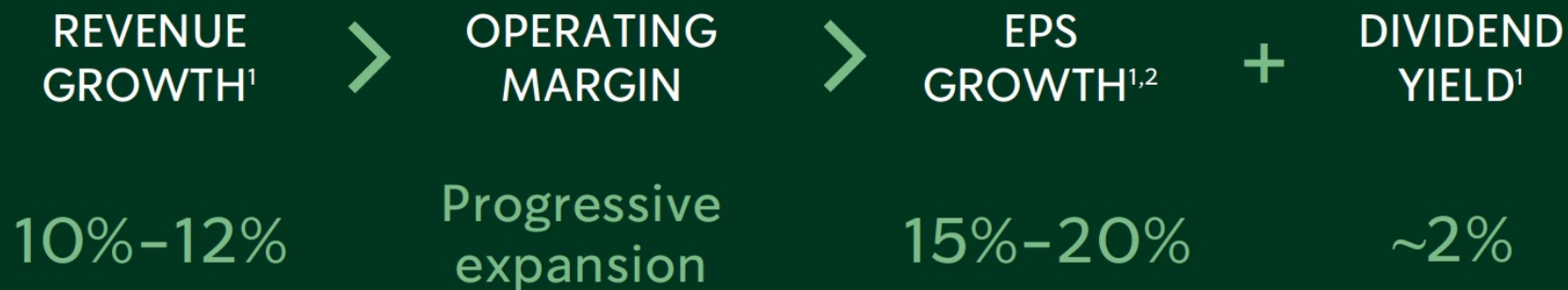
NEW ERA OF GROWTH

EPS 15% – 20%¹



¹Estimated annual growth of non-GAAP EPS for FY23 through FY25. Please refer to "Forward-Looking Statements and Non-GAAP Financial Measures" on our IR website, Starbucks 2022 Investor Day event page, for our statement on the reconciliation of forward-looking non-GAAP estimates.

Adding it all up



REINVENTING TO ACCELERATE GROWTH

¹ Figures are estimated and apply to FY23 through FY25.

² Estimated non-GAAP figure. Please refer to "Forward-Looking Statements and Non-GAAP Financial Measures" on our IR website, Starbucks 2022 Investor Day event page, for our statement on the reconciliation of forward-looking non-GAAP estimates.

15.1%

-270bps YoY²

+8%

U.S. +**12%**
+12% North America

CHINA -**24%**
-9% International

GUIDANCE FISCAL 2023

COMP GROWTH

Global	Near the high end of 7% – 9%
U.S.	7% – 9%
China	Outsized

NEW STORE GROWTH

Global	~7%
U.S.	~3%
China	~13%

GLOBAL REVENUE

10% – 12%

OPERATING MARGIN

Solid Margin Expansion

CapEx

~\$2.5B

INTEREST EXPENSE

\$540M – \$560M

GAAP & Non-GAAP TAX RATE³

Mid-20s

FULL YEAR EPS⁴

GAAP: High End of 15%–20%
Non-GAAP: Low End of 15%–20%

Starbucks 4Q'22 earnings announcement disclosures, Nov. 3, 2022: reports 4Q and FY22 results + FY23 guidance.

Q3 FY26 Outlook

Revenue	\$54.0 billion , plus or minus 2% Does not assume any H2O shipments to China If geopolitical issues recede, could ship \$2 to \$5 billion dollars in H2O revenue in Q3
Gross Margins	73.3% GAAP and 73.5% non-GAAP, plus or minus 50 basis points Continue to expect to exit the year with non-GAAP gross margins in the mid-70% range
Operating Expense	Approximately \$5.9 billion GAAP and \$4.2 billion non-GAAP Expect full year fiscal 2026 operating expense growth to be in the high-30% range
Other Income & Expense	Income of approximately \$500 million for GAAP and non-GAAP Excluding gains and losses from non-marketable and publicly-held equity securities
Tax Rate	16.5% GAAP and non-GAAP, plus or minus 1%, excluding discrete items

Refer to Appendix for reconciliation of Non-GAAP measures.

NON-GAAP GLOBAL OPERATING MARGIN¹

16.1%
+100bps YoY

NON-GAAP EPS¹

\$3.54
+20% YoY

FY24 GUIDANCE

COMP GROWTH

Global	5% – 7%
U.S.	5% – 7%
China	Higher In Q1, 4% – 6% In subsequent quarters

NEW STORE GROWTH

Global	~7%
U.S.	~4%
China	~13%

GLOBAL REVENUE

Low end of 10% – 12%

OPERATING MARGIN

Progressive expansion

CAPITAL EXPENDITURES

~\$3B

GAAP & Non-GAAP TAX RATE²

Mid-20s; higher than FY23

FULL YEAR EPS³

GAAP: 15%–20%

Non-GAAP: 15%–20%

11.2.23 (4Q'23 earnings)

As of the company's Q2 FY24 Earnings Call (4/30/2024)

Fiscal Year 2024 Guidance	
Global Comparable Store Sales Growth	Low single digit decline to flat
U.S. Comparable Store Sales Growth	Low single digit decline to flat
China Comparable Store Sales Growth	Single digit decline
Global Store Growth	~6%
U.S. Store Growth	~4%
China Store Growth	~12%
Global Revenue	Low single digits
Global Operating Margin	Approximately flat
Capital Expenditures	~\$3B
GAAP and non-GAAP Tax Rates	Mid-20s; higher than FY23
GAAP EPS	Flat to low single digits
Non-GAAP EPS ⁽¹⁾	Flat to low single digits

NasdaqGS - Nasdaq Real Time Price • USD

Starbucks Corporation (SBUX)

☆ Follow

↔ Compare

98.57 -0.71 (-0.72%) **98.60** +0.03 (+0.03%)

At close: September 13 at 4:00 PM EDT

After hours: September 13 at 7:58 PM EDT

1D 5D 3M 6M YTD 1Y 5Y All

6.68%

Key Events



2Q'24 earnings
including guidance
downgrade

Announcement of CEO transition
(Niccol announced as new CEO;
Narasimhan steps down).

SBUX stock price March through September, 2024.

The Starbucks Turnaround That Has Baristas and Customers Steamed

Improved service, better seating and heartfelt messages are at the heart of Brian Niccol's turnaround plan, but baristas say they're under pressure.



The Strawberry Matcha Strato Frappuccino, center, at Starbucks requires six ingredients and two blenders to make. Nico Schinco for The New York Times

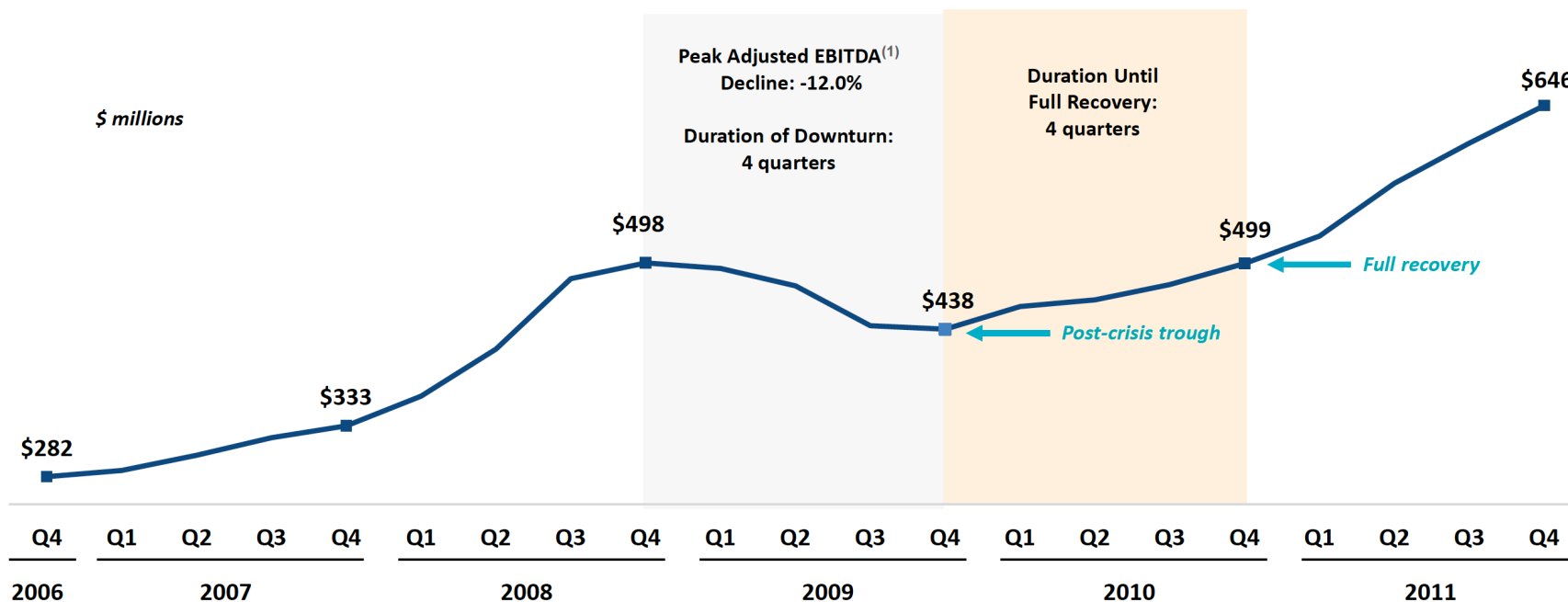
Hotels and airlines; regardless of how little guests, the organisation still incurs significant fixed costs

Operating leverage: extent to which an operation's costs are fixed versus variable—high operating leverage refers to businesses where most costs are fixed, like steel producers, airlines, and hotels. Supermarkets have low operating leverage (costs mostly variable).

- Businesses with high operating leverage are inherently riskier because they are vulnerable to declines in demand (“cyclical” businesses) → higher asset β s.
- This is exactly what we saw with COVID-19.

Performance in a Downturn

Semi-variable cost structure limits financial downside during a downturn



Note: figures show LTM Adjusted EBITDA.

(1) Non-GAAP Measures; see Appendix for definitions and reconciliations to the most directly comparable GAAP financial measures.

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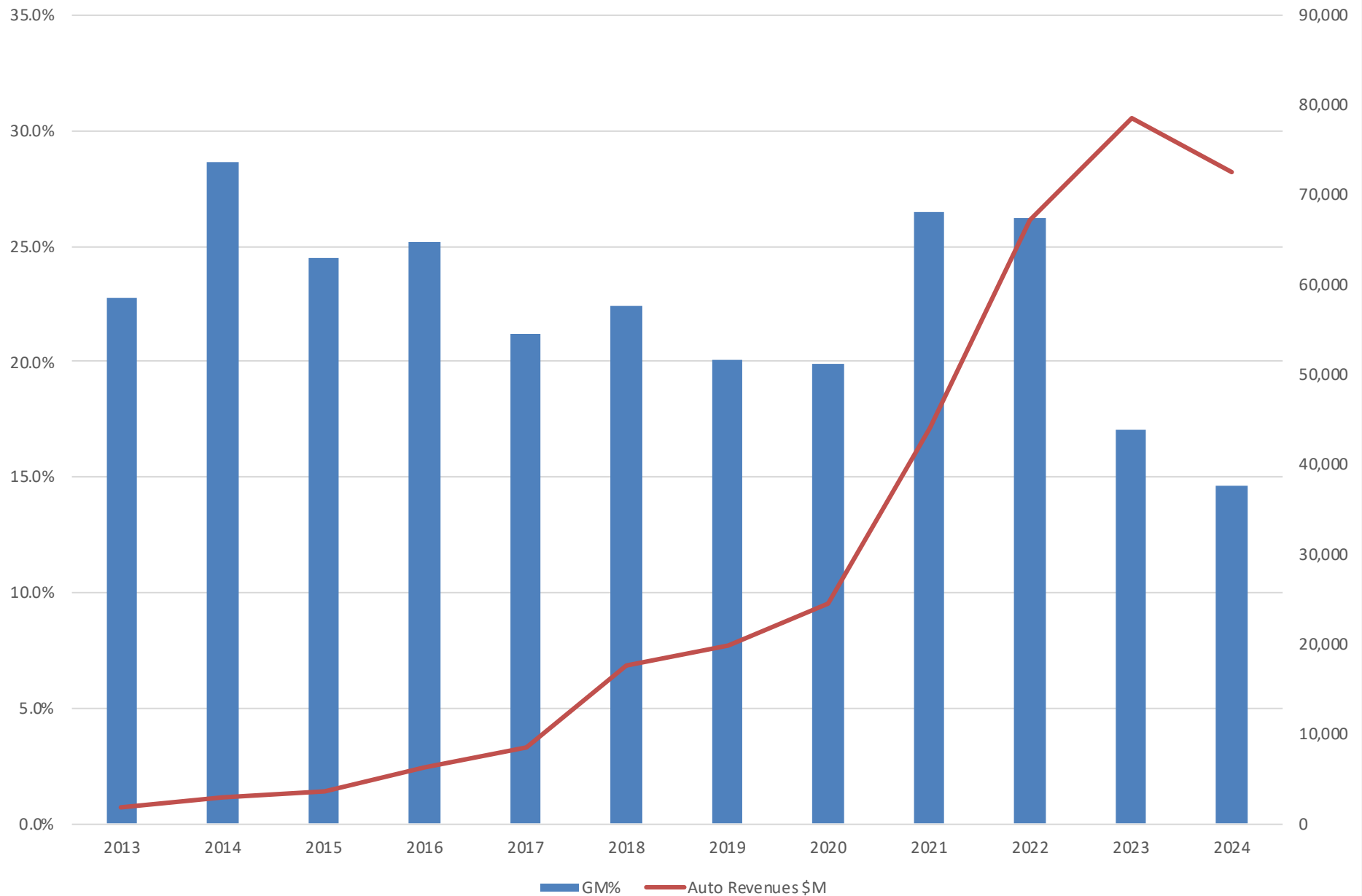
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$$\beta_E = \beta_A + \frac{D}{E}(\beta_A - \beta_D)$$

Asset beta is driven by the operating lev of the busiess

Cost of Capacity, Manufacturing Costs, and Absorption Costing

TSLA Revenues and GM% (Auto business), FY13 through FY24.



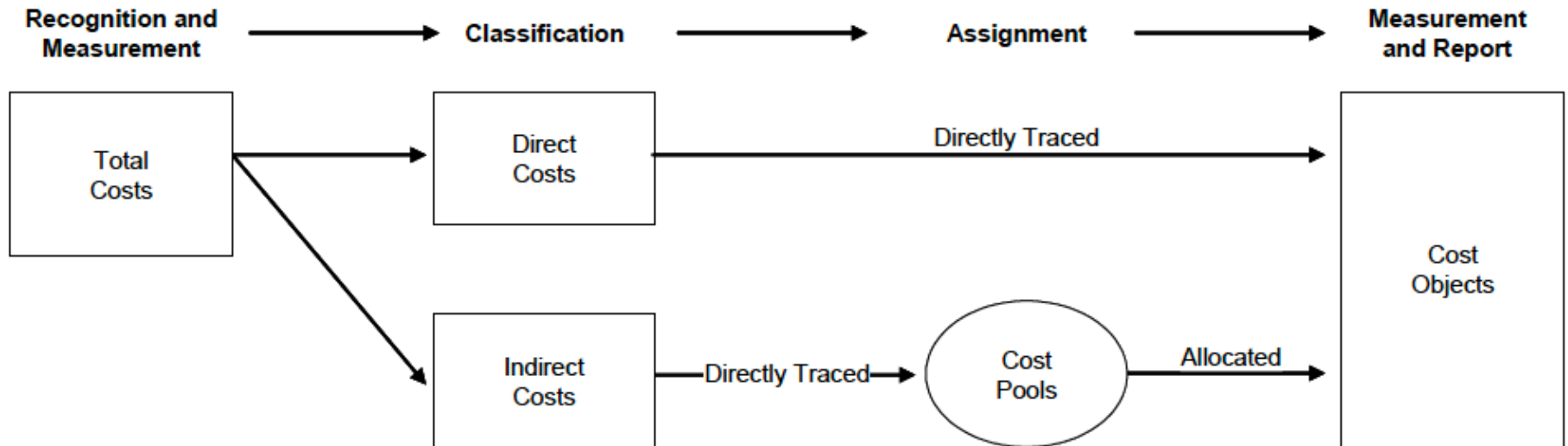
Source: Company 10-K filings (numbers may not be fully comparable year-to-year)

Cost systems

Cost systems perform two functions:

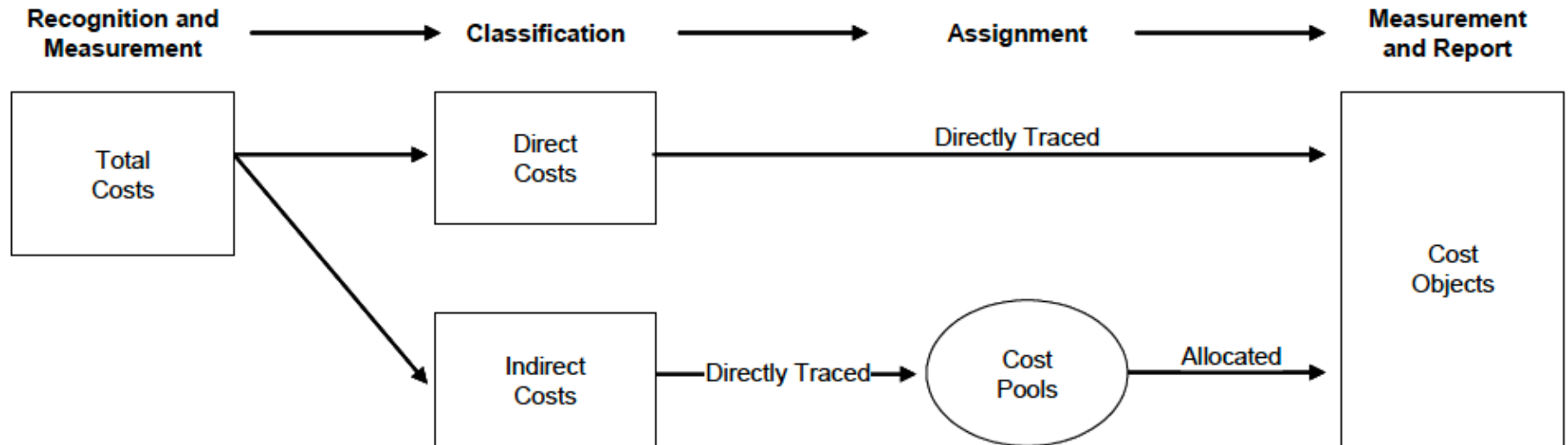
1. **Accumulate cost data:** as resources are acquired, the accounting system captures the associated costs (at some point there is a cash outlay).
2. **Assign costs** to those activities, products, and services that create value (**cost objects**).
 - In RegionFly, the routes are the relevant cost objects.
 - Done either **directly** (direct costs) or **indirectly** by way of an **allocation rule** (indirect costs).

Basic structure of Cost Systems.



Source: Introduction to Cost-Accounting Systems (HBS).

Basic structure of Cost Systems.



Whether and how to allocate indirect costs to cost objects is always **purpose-dependent** – we don't always allocate indirect costs and may do so in different ways for different purposes (e.g., performance evaluation versus control).

Costs and resources

Organizations use two types of resources:

- **Consumable resources:** resources are consumable if the cost depends on how much is used (consumed). This cost will often be a *variable cost*, such as the cost of materials (e.g., fuel for American Airlines).

Costs and resources

Organizations use two types of resources:

- **Consumable resources:** resources are consumable if the cost depends on how much is used. This cost will often be a *variable cost*, such as the cost of materials (e.g., fuel for American Airlines).
- **Capacity-related resources:** Cost of supplying capacity depends on the amount of capacity the entity makes available, not how much is used. This cost will often be *fixed* (e.g., the cost to Tesla of keeping its Fremont plant open each month; the cost of keeping American's current aircraft fleet available to fly passengers each quarter).

Current Installed Annual Vehicle Capacity

Region	Model	Capacity	Status
California	Model S / Model X	100,000	Production
	Model 3 / Model Y	>550,000	Production
Shanghai	Model 3 / Model Y	>950,000	Production
Berlin	Model Y	>375,000	Production
Texas	Model Y	>250,000	Production
	Cybertruck	>125,000	Production
Nevada	Tesla Semi	-	Pilot production
Various	Next Gen Platform	-	In development
TBD	Roadster	-	In development

Total capacity > 2.3M vehicles

Source: TSLA 2Q'25 earnings deck (unchanged from 2Q'24).

OPERATIONAL SUMMARY
(Unaudited)

	Q2-2024	Q3-2024	Q4-2024	Q1-2025	Q2-2025	YoY
Model 3/Y production	386,576	443,668	436,718	345,454	396,835	3%
Other models production	24,255	26,128	22,727	17,161	13,409	-45%
Total production	410,831	469,796	459,445	362,615	410,244	0%
Model 3/Y deliveries	422,405	439,975	471,930	323,800	373,728	-12%
Other models deliveries	21,551	22,915	23,640	12,881	10,394	-52%
Total deliveries	443,956	462,890	495,570	336,681	384,122	-13%
of which subject to operating lease accounting	10,227	14,449	26,962	13,721	6,670	-35%
Total end of quarter operating lease vehicle count	171,353	168,867	180,523	179,930	172,882	1%
Global vehicle inventory (days of supply) ⁽¹⁾	18	19	12	22	24	33%
Storage deployed (GWh)	9.4	6.9	11.0	10.4	9.6	2%
Tesla locations	1,286	1,306	1,359	1,390	1,454	13%
Mobile service fleet	1,896	1,933	1,895	1,799	1,684	-11%
Supercharger stations	6,473	6,706	6,975	7,131	7,377	14%
Supercharger connectors	59,596	62,421	65,495	67,316	70,228	18%

Total production TTM = 1.70M vehicles (deliveries of 1.68M)

Production capacity > production, so need to consider cost of capacity being supplied.

Source: 2Q'25 earnings deck.

Reference Daikin example.

$$\begin{array}{l} \text{Cost of} \\ \text{Resources} \\ \text{Supplied} \end{array} = \begin{array}{l} \text{Cost of} \\ \text{Resources} \\ \text{Used} \end{array} + \begin{array}{l} \text{Cost of} \\ \text{Unused} \\ \text{Capacity} \end{array}$$

Why? this is considered fixed costs.

$$\begin{array}{ccccc} \text{Cost of} & & \text{Cost of} & & \text{Cost of} \\ \text{Resources} & = & \text{Resources} & + & \text{Unused} \\ \text{Supplied} & & \text{Used} & & \text{Capacity} \end{array}$$

Remember me!!





Costs and resources

Costs are direct or indirect:

- Costs are **direct** if we can unambiguously assign them to a specific cost object, such as a product or service (e.g., the cost of raw materials – carbon fiber, aluminum, high strength steel – used to produce a Lamborghini). The costs of *consumable resources are usually direct costs*.

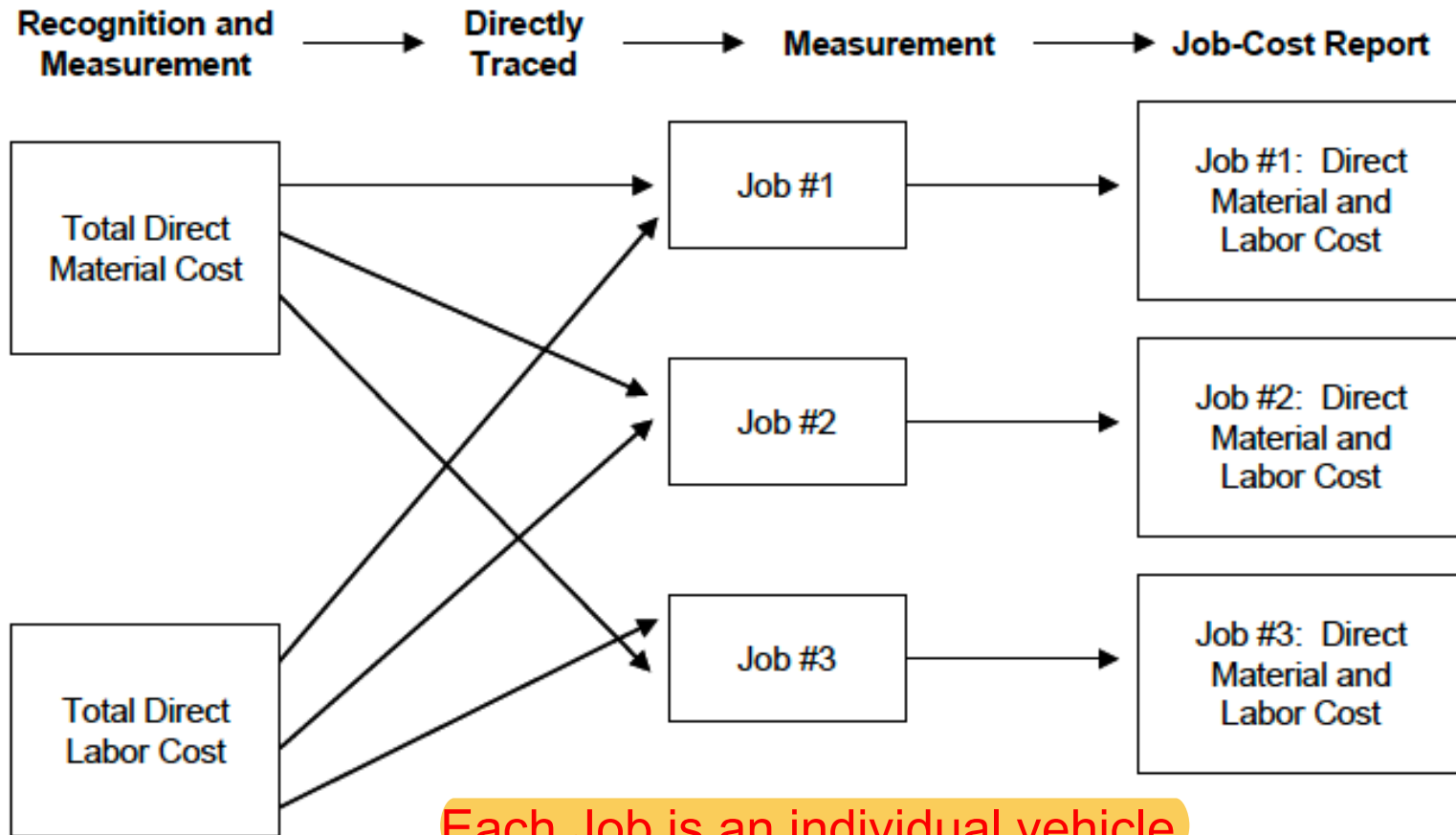
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*But not a Toyota (when production volumes are high and products homogeneous, raw materials and direct labor may be allocated to inventory cost pools).

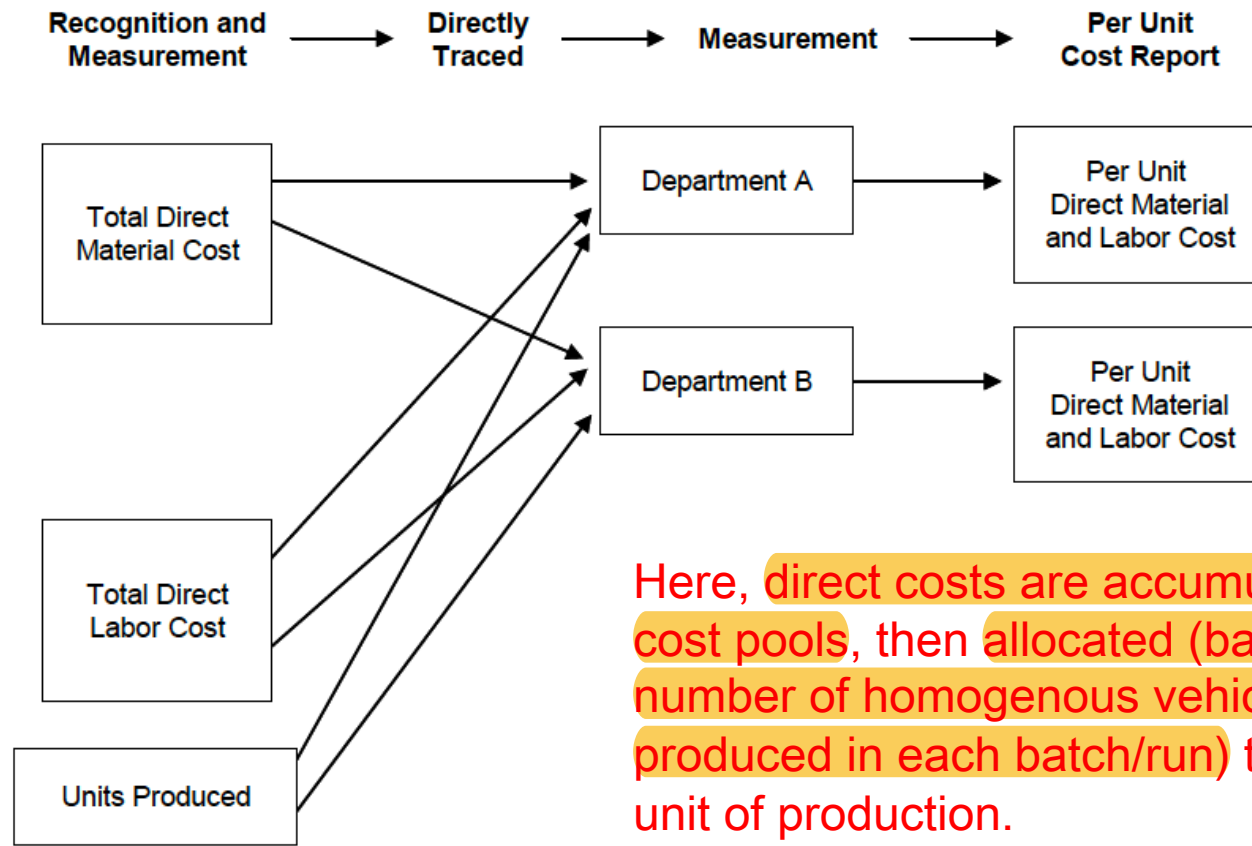
Job-Order Cost System (direct costs only): relatively small number of identifiable units of product output, as with a Lamborghini.



Each Job is an individual vehicle.

Source: Introduction to Cost-Accounting Systems (HBS).

Process Cost System (direct costs only): large numbers of relatively homogeneous units (can be some variation), as with a Toyota.



Here, direct costs are accumulated in cost pools, then allocated (based on number of homogenous vehicles produced in each batch/run) to each unit of production.

Source: Introduction to Cost-Accounting Systems (HBS).

Costs and resources

Costs are direct or indirect:

- Costs are **direct** if we can unambiguously assign them to a specific cost object, such as a product or service (e.g., the cost of components used to build a Lamborghini*). The costs of consumable resources are usually direct costs.
- All other costs are **indirect**. Most capacity-related resource costs are indirect (for example, the cost of keeping a plant running to produce a variety of Tesla models and variants).

Are all variable costs also direct costs, and all fixed costs also indirect costs?

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	Direct Cost	Indirect Cost
Variable Cost	Cost of components purchased from supplier (e.g., OLED screens for iPhones)	Cost of electrical power to run machines at plant that produces multiple products
Fixed Cost	Cost of rent on facility that only assembles iPhone 16 Pro (cost object is entire product line).*	Cost of rent on facility that assembles several iPhone models (cost object is specific model).*

*The same cost can be a direct or indirect – it *depends on the cost object*.

Tesla, Inc.
Consolidated Statements of Operations
(in millions, except per share data)

	Year Ended December 31,		
	2024	2023	2022
Revenues			
Automotive sales	\$ 72,480	\$ 78,509	\$ 67,210
Automotive regulatory credits	2,763	1,790	1,776
Automotive leasing	1,827	2,120	2,476
Total automotive revenues	77,070	82,419	71,462
Energy generation and storage	10,086	6,035	3,909
Services and other	10,534	8,319	6,091
Total revenues	97,690	96,773	81,462
Cost of revenues			
Automotive sales	61,870	65,121	49,599
Automotive leasing	1,003	1,268	1,509
Total automotive cost of revenues	62,873	66,389	51,108
Energy generation and storage	7,446	4,894	3,621
Services and other	9,921	7,830	5,880
Total cost of revenues	80,240	79,113	60,609
Gross profit	17,450	17,660	20,853
Operating expenses			
Research and development	4,540	3,969	3,075
Selling, general and administrative	5,150	4,800	3,946
Restructuring and other	684	—	176
Total operating expenses	10,374	8,769	7,197
Income from operations	7,076	8,891	13,656

Gross margin (automotive) based on above is 14.6% in 2024, 17.1% in 2023, and 26.2% in 2022.

Cost of Revenues

Automotive Segment

Automotive Sales

Cost of automotive sales revenue includes direct and indirect materials, labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, shipping and logistic costs, vehicle connectivity costs, FSD (Supervised) ongoing maintenance costs, electricity costs for our free Supercharging programs and reserves for estimated warranty expenses. Cost of automotive sales revenues also includes adjustments to warranty expense and charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of automotive sales revenue benefits from manufacturing credits earned, amounting to \$625 million and \$359 million for the years ended December 31, 2024 and 2023, respectively.

Source: TSLA 2024 10-K.

Cost of automotive sales revenue includes **direct and indirect materials, labor costs, manufacturing overhead, including depreciation costs of tooling and machinery**, shipping and logistic costs, vehicle connectivity costs, FSD (Supervised) ongoing maintenance costs, electricity costs for our free Supercharging programs and reserves for estimated warranty expenses... also includes adjustments to warranty expense and **charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value** and to provide for obsolete and on-hand inventory in excess of forecasted demand.

Includes more than manufacturing costs. Manufacturing costs based on absorption costing (written down to NRV if necessary under lower-of-cost-or-market rule).

Manufacturing Costs

- Financial accounting distinguishes **product costs** (costs assigned to product) from **period costs** (costs not directly linked to products, which are expensed as incurred).

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- GAAP (including IFRS) requires **Absorption Costing** to measure product costs for manufacturing.

We classify manufacturing costs into three buckets:

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3. **Manufacturing overhead:**
 - a) Indirect Labor: cost of workers who don't work directly on product but are necessary for plant to operate.
 - b) Indirect Materials: cost of materials that are not part of the finished product (e.g., lubricants for machinery).
 - c) Other manufacturing costs, such as utilities costs and depreciation of plant and equipment.

Manufacturing Costs

Manufacturing overhead (MOH) is separated into **fixed manufacturing overhead** (overhead costs that do not vary with production activity) and **variable manufacturing overhead** (overhead costs that vary with the level of plant activity).

Both are *allocated to product* under **Absorption Costing** (GAAP) – i.e., **they are product costs.**

Manufacturing Costs

Under GAAP/IFRS and the US Tax Code, all manufacturing costs, including fixed overhead costs, are product costs and so are allocated to inventory.

- For most applications, *not a useful economic approach* to measuring inventory costs because the allocation is inherently arbitrary (does not reflect what causes costs to vary).
- Known as **full costing** or **absorption costing**. (Alternative is a **direct costing** or **variable costing**, under which only direct, variable manufacturing costs are assigned to products.)

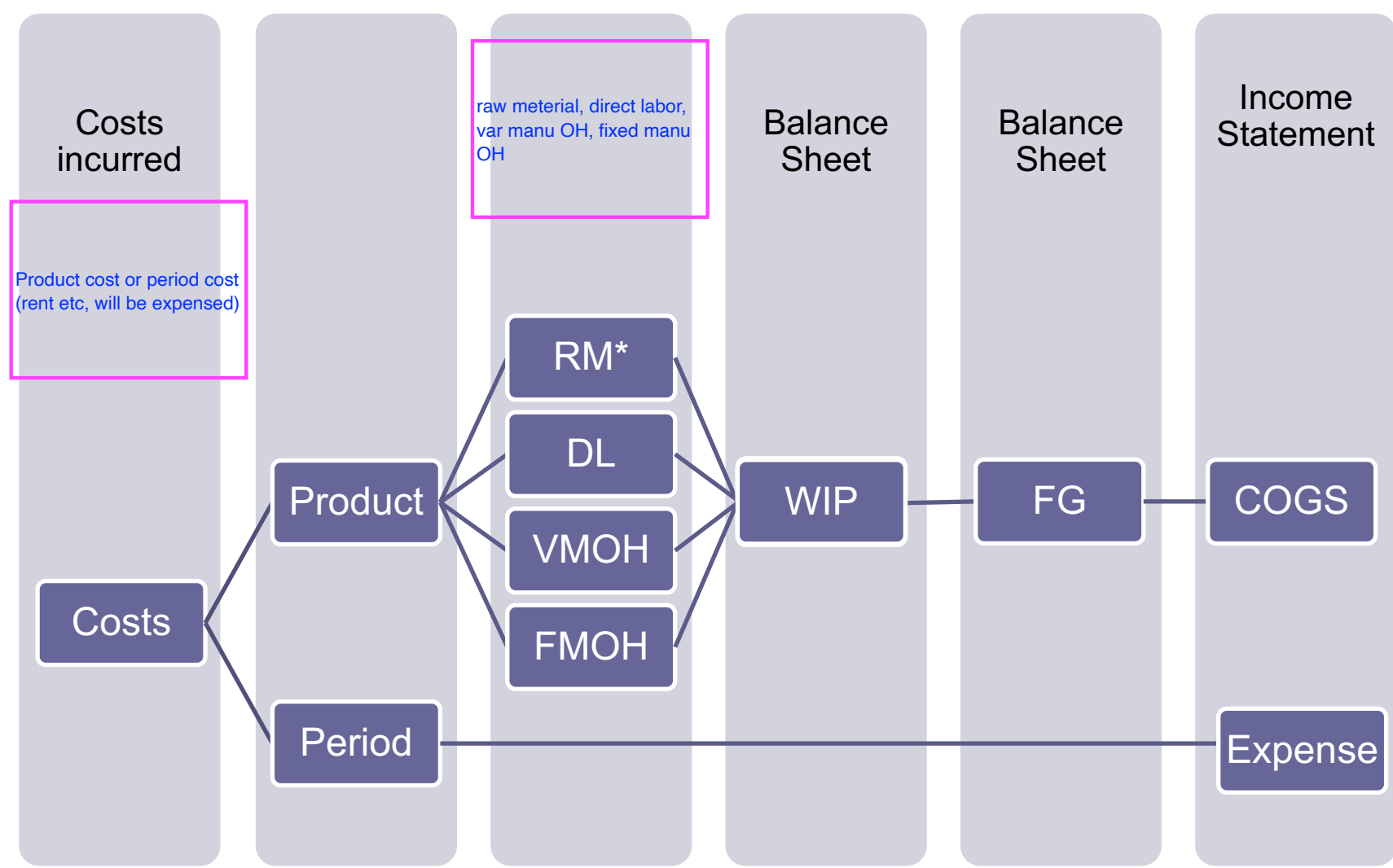
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Remember me!!

Manufacturing cost flows (GAAP):



*RM is an inventory account when purchased, before used in production process.

ASC 330 (US GAAP) Inventory Costing

- Inventories measured at cost = acquisition and production costs under absorption costing (subject to *lower of cost-or-market rule*: write down to NRV if $<$ cost).
- Variable MOH costs are allocated to product based on ***actual production levels***:
 - GAAP relies on on *past* transactions and events.
 - must wait until end of period to calculate inventory cost – so cannot use for budgeting or planning.

ASC 330 Inventory Costing

- Fixed MOH costs allocated to product based on “***normal capacity***” as defined under GAAP:

“range of production levels... expected to be achieved over a number of periods or seasons under normal circumstances... judgment is required.”
- Goal is to avoid period-to-period variation in inventory costs due to fluctuations in actual or planned production.
- Requires management judgment.

Capacity measures/terms:

Plant (maximum) capacity or output = 100,000

“Normal Capacity” (GAAP) = 95,000

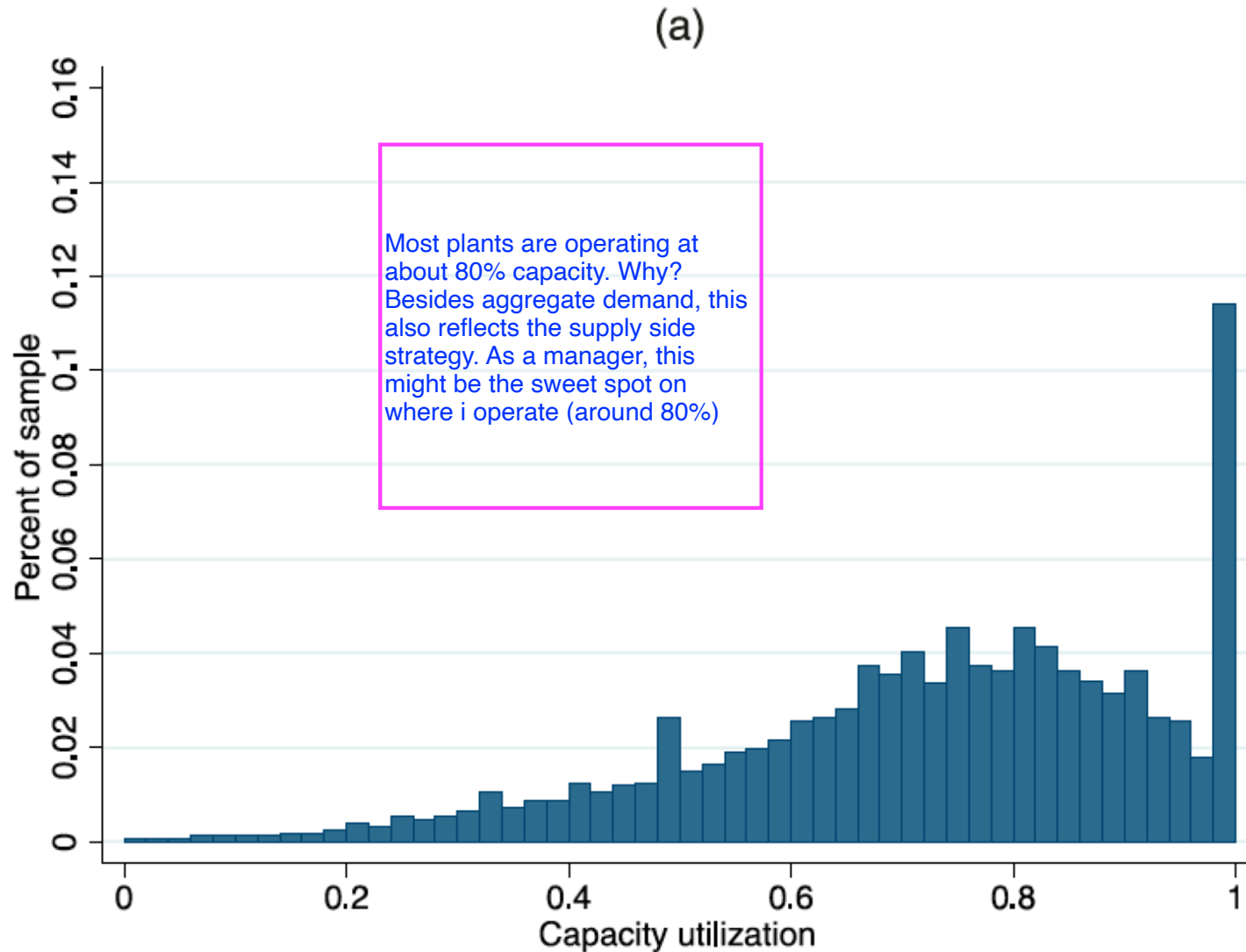
Actual production in 2024 = 87,478

Practical capacity = 90,000

Focus on practical
capacity

Practical capacity is typically < maximum capacity because marginal production costs usually increase as we get close to the maximum. Also, most plants operate with a certain amount of downtime (planned and unplanned).

Figure 2. (Color online) Histogram of Plant Capacity Utilization over All Years (Full Sample)



Source: Ederhofer, Nagar, Rajan (2021, Management Science).

$$\text{FMOH/unit} = \frac{\text{Total FMOH for period}}{\text{Normal capacity}}$$

$$\text{VMOH/unit} = \frac{\text{Total VMOH for period}}{\text{Actual production}}$$

Under GAAP/absorption costing, these costs are added to the direct costs of the product.

For internal decisions (budgeting, performance evaluation, pricing) would normally:

- a) Exclude FMOH from inventory cost, and
- b) Compute VMOH/unit using *practical capacity*, not actual production.

Summary of inventory costing under absorption costing vs. for internal purposes:

Components of inventory costs:	Absorption costing (GAAP)	Internal decisions, performance evaluation, control
Fixed MOH/unit	Compute using Normal Capacity	Do not allocate
Variable MOH/unit	Compute using Actual Production	Compute using Practical Capacity or do not allocate
Direct material and Direct Labor per unit	Include (variable costs)	Include (variable costs)
Total Unit Costs	Inventory cost per unit for GAAP/IFRS Reporting	Inventory cost per unit for internal decision-making and control

Most crucial element in management to understand this portion.

$$\begin{array}{l} \text{Cost of} \\ \text{Resources} \\ \text{Supplied} \end{array} = \begin{array}{l} \text{Cost of} \\ \text{Resources} \\ \text{Used} \end{array} + \begin{array}{l} \text{Cost of} \\ \text{Unused} \\ \text{Capacity} \end{array}$$

↑
Allocate to product
using *practical*
capacity.

↑
Don't allocate to
product; use for
control

ASC 330 Inventory Costing

- If the plant is not producing to capacity, any fixed overhead not allocated to product is a period cost, normally included in COGS.
- This is the cost of maintaining excess capacity, a useful number economically (but may not be disclosed in GAAP financials).
- Why do we care? If plant operates below capacity, capacity costs are being incurred and need to be evaluated (important for **control**). Also a possible *opportunity cost* (foregone revenues).

Absorption costing and earnings management

Absorption costing gives managers two levers to manage reported gross margin under GAAP:

- Understate the denominator in the allocation rate (“normal capacity”): increases overhead allocated to each unit, reducing COGS if units unsold at the end of the period.*
- If production > sales, more of the period’s MOH costs ends up in inventory on the balance sheet (and not in current period COGS).
 - Increases in production relative to sales increase gross margin. This may happen naturally due to an unforeseen decline in demand or strategically to increase reported GM.

*Fixed MOH only. (Variable MOH uses actual production units as denominator.)

Absorption Costing: Fixed Manufacturing Overhead

Assume \$2M of fixed manufacturing overhead cost for 2024 and no opening inventory

Management sets Normal Capacity = 100,000 Assumption/judgment under GAAP

FMOH/unit = \$20.00

Assume also:

Production for 2024 80,000

Sales for 2024 75,000 Determines booked revenue

What's reported in COGS and Closing Inventory?

Closing inventory = \$100,000 (Production - Sales) x FMOH/unit

COGS (allocated to product) \$1,500,000 Sales x FMOH/unit

Remaining unallocated FMOH \$400,000

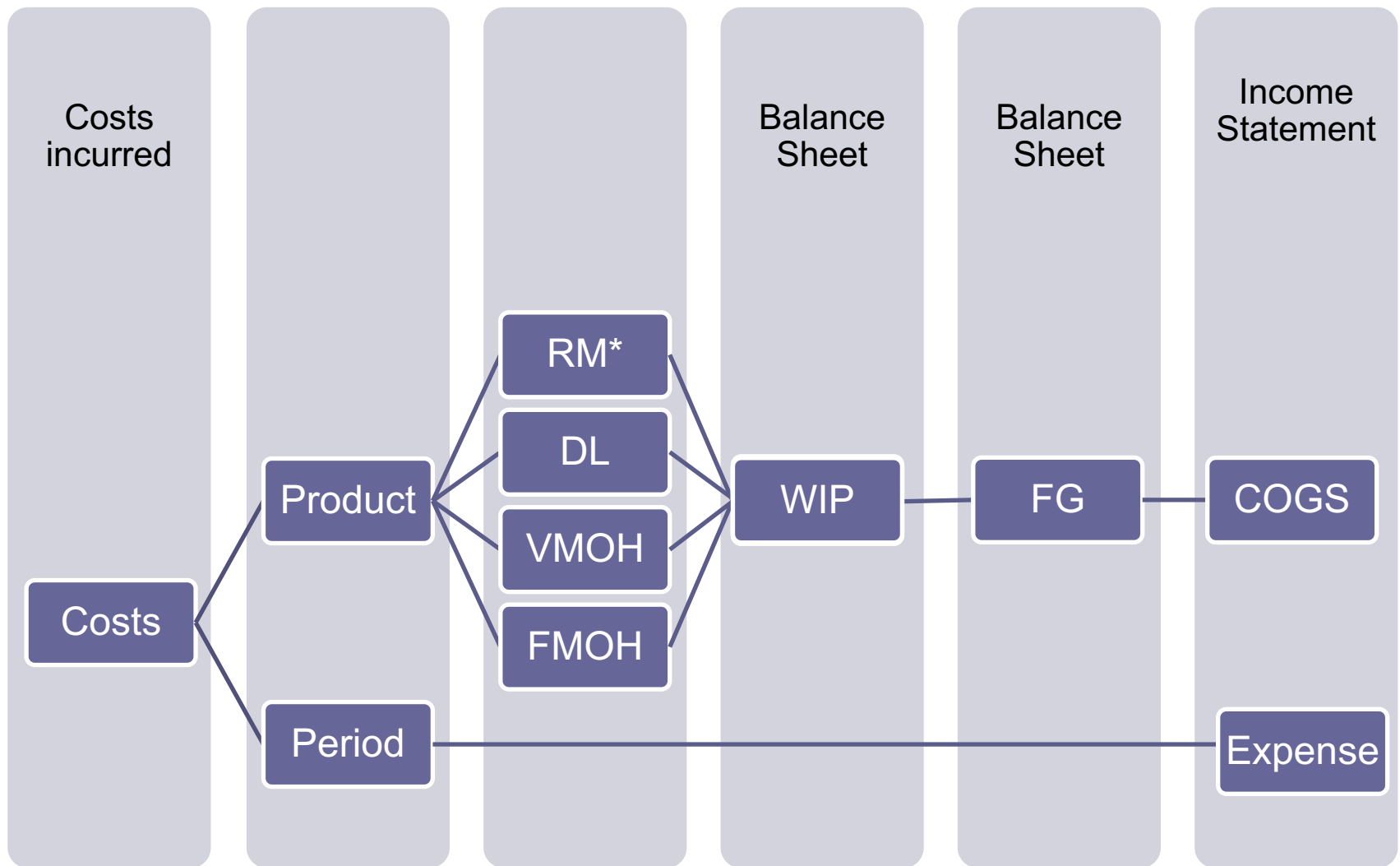
Total COGS = \$1,900,000

Total FMOH \$2,000,000

Total MOH must be fully allocated to COGS and Closing Inventory (i.e., they must sum to \$2M).

(Because there's no opening inventory, production cannot be less than sales.)

Manufacturing cost flows (GAAP):



*RM is an inventory account when purchased, before used in production process.

Adjusted EBITDA margin percentage for the total Company was 28.8% for the three months ended September 30, 2018 compared to 33.5% for the same period in the prior year. The decrease for the three months ended September 30, 2018 compared to the three months



ended September 30, 2017 was driven by the Space Systems segment and is primarily related to an increase in estimated costs to complete programs as a result of supplier performance issues experienced during the third quarter of 2018, under-absorption of fixed indirect overhead due to a reduced number of satellite awards, reduced labor productivity and a recovery of liquidated damages which occurred during 2017 contributed to the decrease in EBITDA. Excluding the GEO Comsat business, adjusted EBITDA margin percentage for the total Company was 38.5% for the three months ended September 30, 2018 compared to 41.9% for the same period in the prior year, and on a year-to-date basis was 40.1% compared to 41.0% in the prior year period.

Designing cost systems

- As illustrated in the VMD MIC case, the goal is to make cost pools as homogeneous as possible, so we do a better job of explaining variation in overhead costs with the activity measure used to allocate costs.
- However, as we increase the number of cost pools we add complexity and cost to our cost system so there is a tradeoff.

Designing cost systems

Which activity measure to use?

efficient and
reasonably accurate
over several years

- **practical capacity**, as opposed to actual production (only observable after the end of the period) or planned/budgeted capacity for a given period.
- **Practical capacity does not usually vary while planned capacity does**, causing variation in unit costs that doesn't reflect variation in the cost/efficiency of production. This is problematic if costs are used for:
 - Performance evaluation (based on costs or gross margin).
 - Pricing (as with demand-induced death spiral in VMD).

Designing cost systems

- Goal is an economically sensible approach to determining cost that will be useful for decisions—both forward- and backward-looking—such as pricing, product mix, and performance evaluation.

Non-manufacturing Costs

SG&A costs and R&D costs. Under GAAP, such costs are treated as **period costs** (expensed as incurred).

Non-manufacturing Costs

SG&A costs and R&D costs. Under GAAP, such costs are treated as period costs (expensed as incurred).

For our purposes – management decisions – the distinction between manufacturing and non-manufacturing costs is artificial, we're interested in **all relevant economic costs** of activities and processes.

- We should assign direct (variable) marketing and selling costs to products to calculate contribution margins, which can inform pricing decisions.
- What about R&D costs?

Knowledge and the Allocation of Decision Rights in Organizations

Markets and Organizations

Markets that work well have strong rule of law to protect this alienability rights.

Alienability: the right to sell/transfer property rights (assets) and capture the proceeds of exchange.

- Alienability (and a property rights/legal system that enforces those rights) is at the core of what makes markets work.
- Mechanism through which markets co-locate decision rights with information.
- Those with the best information value assets the most highly (e.g., real estate, collectible art, NFTs, pro sports franchises).

Markets and Organizations

The absence of alienable decision rights (and markets) within organizations differentiates decisions made **in markets** (and outside firms) from those made **within firms**:

→ Organizations must develop internal systems to allocate scarce resources – need to assign decision rights (i.e., who decides how scarce resources are deployed?).

Hayek and Knowledge

- To be effective, organizations co-locate decision-making authority with the relevant information.
- However, people have limited cognitive capacity: the CEO of Apple cannot make all decisions.
- In addition, people have psychological biases (e.g., averse to losses, overconfident and miscalibrated, attribution biases).
 - Humans are limited in their ability to process information and subject to biases.

Hayek and Knowledge

Specific = Tacit / idiosyncratic knowledge. project quirks, what type of contract staff not to hire.

- Can either move information up to the decision-maker (a centralized organizational structure) or move decision-making down to information (decentralized structure).
- Specific vs. General knowledge. **Specific knowledge** is expensive to transfer and not easily observable by other agents; difficult to aggregate and summarize.
- Walgreens' local store managers know best what products to stock in front of store to meet local customer demand (e.g., how much sunscreen to stock in July).



Walgreens store in downtown Chicago (Gold Coast)



Walgreens store in Des Moines, Iowa.

Hayek and Knowledge

- **General knowledge** is transferable among agents at low cost or easily observable by other agents
 - Prices and quantities are good examples (the cost of a generic drug product purchased centrally can easily be communicated to all store managers).
- Specific knowledge also known as **tacit knowledge**; Hayek refers to this as knowledge of “particular circumstances of time and place.” **Idiosyncratic** knowledge is a type of specific knowledge.

HOW UNITED AIRLINES WILL EMPOWER GATE AGENTS WITH MORE DISCRETION

👤 MATTHEW KLINT / 🕒 SEPTEMBER 23, 2017 / 💬 10 COMMENTS



Airlines tend to oversell flights. It does not hurt the airline financially. But reputation damages occur when we have to take people off the full flight. Example; doctor Dao is being taken off the flight even though he's already sitting on the plane seat.



The **Dao incident** was a **life-changing event** at United. While the airline may not have been **hurt financially** by the unforced error, it is causing a change in culture.

Hayek and Knowledge

Gate agent possess a specific knowledge of which customer to remove from the flight (they understand the local context, the customer profile and behavior)

- Why do airlines empower gate agents to make decisions about specific flights (e.g., overbooking)?
- To make the right decision, need to know how many customers show up at the gate and the compensation they're willing to take in exchange for taking a later flight – knowledge of a “particular circumstances of time and place” that is costly to transfer to a centralized decision-maker – i.e., *specific knowledge*.
- Airlines now use apps powered by AI/ML to gather and process specific knowledge.

11:03

5G+ 100



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[Terms and conditions](#)

Continue with offer

This tech is meant to extract the WTP from each customer.
Then the algo will determine who is selected.

In sum, for organizations to be effective, we need to:

- Figure out how to allocate decision-rights so decision-makers have access to best information. The type of knowledge (general vs. specific) will be important here.
- But there is a tradeoff – if we push decision-making to lower levels of the organization, we lose control over how resources are deployed.
- Design a control system that:
 - (a) measures and evaluates performance, and
 - (b) provides rewards for agents (promotion, compensation) that reflects and so motivates performance/good decisions.

A few words on Midwest Office Products

Midwest Office Products.

Steps (time-driven ABC):

1. Identify activities/processes necessary to fulfill orders.
2. Figure out the total cost of each.
3. Estimate **practical capacity** of the activities/processes, thinking carefully about the appropriate cost driver (e.g., is it number of cartons or time?).
4. Calculate cost rate of supplying capacity.
5. Assign costs to orders based on extent to which each consumes capacity of the different activities/processes.